

# johnny\_4\_docs

System-level documentation for the Johnny 4 robot: everything that spans more than one board repo.

## Contents

| File                                        | What it is                                                                                                                         |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">POWER.md</a>                    | Power distribution for the robot side: which pin each board takes 5V on, buck converter zones, USB flashing/backfeed rules, budget |
| <a href="#">j4_master_pin_diagram.pdf</a>   | All seven boards' pin diagrams, one full letter page each, printable as one document                                               |
| <a href="#">tools/gen_master_pin_pdf.py</a> | Regenerates the master PDF (and the standalone j4_controller pin diagram) from the board repos' READMEs                            |

## Regenerating the master pin diagram

The generator parses each repo README's `## Pin diagram` fenced block and `## Pin assignments` table, so the PDF always matches the repos. After changing any pinout, update that repo's README and run:

```
python tools/gen_master_pin_pdf.py
```

Requires `reportlab`. The script expects the board repos checked out as siblings of this repo ( `../j4_controller`, `../j4_receiver`, and so on).

## The board repos

Signal chain: [j4\\_controller](#) --ESP-NOW--> [j4\\_receiver](#) --UART--> [j4\\_stepper\\_neck](#) --UART--> [j4\\_stepper\\_eyes](#), with [j4\\_talk](#) on the receiver's second UART and [j4\\_display\\_left](#) + [j4\\_display\\_right](#) on the controller's UARTs. Printable parts live in [johnny\\_4\\_3D\\_printable\\_parts](#).